ASE CHAPTER 2

T5 Apply & Scenario Selection Pack · Life-Cycle Models

Exercise 13 (Overview): Engineering Principles of Life-Cycle Model Selection

Selecting an SDLC model is an engineering decision governed by evaluating four interrelated project parameters rather than universal preference: 1. **Requirements Certainty:** Are requirements fully understood and fixed upfront, or fluid, evolving, and unclear? 2. **Technical and Project Risk:** Does the project rely on unproven technologies, tight hardware constraints, or complex distributed integrations? 3. **User Involvement Profile:** Is the customer available for continuous daily feedback, or restricted to formal milestone approvals? 4. **Development Team and Delivery Constraints:** Does the team possess deep domain expertise, and are delivery deadlines aggressive (60–90 days) requiring parallel modular development?

Exercise 13.i: Which life-cycle model would you follow for developing software for: A Game? Justify your selection with an appropriate reason.

Recommended Model: Prototyping Model (or Evolutionary Prototyping / Agile)

Engineering Justification: - **Extreme User Experience and Interface Uncertainty:** Video games depend heavily on subjective human factors—gameplay mechanics, visual appeal, user immersion, graphical responsiveness, and intuitive controls—which cannot be captured in a static upfront SRS document. - **Exploratory Playtesting and Refinement: The prototyping paradigm enables rapid development of working gameplay mock-ups.** These prototypes are evaluated directly through user playtesting to tune controls, adjust difficulty pacing, and eliminate disorienting interface flaws before committing to full-scale graphic asset production and rendering engine architecture. - **Technical Risk Resolution:** Allows developers to test 3D rendering framerates, physics calculations, and hardware compatibility on target consoles before locking down the design.

> **ملاحظة الطالب والفخ الامتحاني:** - ألعاب الفيديو تعتمد بالكامل على تجربة المستخدم (Gameplay & UX & GUI)؛ لا يمكن كتابة مواصفات جامدة لها مسبقاً في الشلال. البروتوتايب هو الخيار الهندسي الأمثل لاختبار متعة اللعبة وضبط تفاعلها.

Sources: B.B. Agarwal et al. (2010), §2.3, pp. 41–44; §2.8, p. 51 (Table 2.2); K.K. Aggarwal & Singh (2007) Slides 37– 40.

Exercise 13.ii: Which life-cycle model would you follow for developing software for: A New Text Editor? Justify your selection with an appropriate reason.

Recommended Model: Iterative Enhancement Model

Engineering Justification: - **Clear, Stratifiable Core Functionality:** Text editors possess a well-understood, universally recognized functional core: basic character input, file buffer loading, saving, cut-copy-paste, and document printing. - **Incremental Expansion of Capability: The iterative enhancement model is ideally suited because software can be partitioned into distinct increments delivered sequentially to users (matching the canonical example in B.B. Agarwal et al., pp. 47–48):** - **Increment 1 (Core Product): Basic file management, editing, and document production functions.** - **Increment 2: More sophisticated editing and document production capabilities.** - **Increment 3: Spelling and grammar checking capabilities.** - **Increment 4: Advanced page layout capability.** - **Immediate Utility and Feedback:** Users obtain a functional, lightweight core product early on, while subsequent releases enhance advanced capabilities based on actual operational feedback.

ملاحظة الطالب والفخ الامتحاني: - محرر النصوص هو المثال الحرفي الذي وضعه كتاب أغروال ص.47–48 لشرح Iterative Enhancement Model. احرص على ذكر النواة الأساسية أولاً ثم الإضافات التدريجية في كل إصدار.

Sources: B.B. Agarwal et al. (2010), §2.6, pp. 47–48; §2.8, p. 51 (Table 2.2).

Exercise 13.iii: Which life-cycle model would you follow for developing software for: A Compiler for a New Language? Justify your selection with an appropriate reason.

Recommended Model: Waterfall Model (Linear Sequential Model) — Conditional Recommendation

Engineering Justification & Explicit Assumptions: - **Contextual Assumption (Stable Language Definition):** The exercise statement does not specify whether the language specification is stable or evolving. Assuming the formal grammar, lexical syntax, semantics, and target instruction set are established and stable upfront: - **Alignment with Selection Matrices:** According to the Requirements Characteristics selection matrix (K.K. Aggarwal & Singh, Slide 37), the Waterfall model is recommended when **"requirements are easily understandable and defined early"** and **"can we define requirements early in the cycle: Yes"**. - **Sequential Architectural Pipeline:** Classic compiler architecture follows an unbranching pipeline of successive phases: **Lexical Analysis (Scanning)** → **Syntax Analysis (Parsing)** → **Semantic Analysis** → **Intermediate Code Generation** → **Code Optimization** → **Code Generation.** - **Alternative Condition (Experimental / Evolving Language):** If the programming language grammar or type system is experimental and evolving, the Waterfall model is unsuitable because changes to syntax invalidate parser tables and AST definitions across downstream passes; an Evolutionary or Incremental development approach would be required instead.

ملاحظة الطالب والفخ الامتحاني: - نص السؤال لم يحدد حالة المتطلبات؛ لذلك اجعل إجابتك في الامتحان مشروطة بافتراض هندسي صريح: - إذا افترضنا أن مواصفات وقواعد اللغة مستقرة ومفهومة مسبقاً ← نموذج الشلال (Waterfall) خيار قياسي ومعقول ومطابق لمصفوفة الاختيار (Slide 37) نظراً لخطية مراحل بناء المترجم وغياب غموض الواجهات. - إذا كانت اللغة تجريبية وقواعدها تتغير باستمرار ← يفشل الشلال ويتطلب الأمر نموذجاً تطورياً أو تكرارياً.

Sources: B.B. Agarwal et al. (2010), §2.2, pp. 36–41; §2.8, p. 51 (Table 2.2); K.K. Aggarwal & Singh (2007) Slide 37.

Exercise 13.iv: Which life-cycle model would you follow for developing software for: A Software for Hospital Management? Justify your selection with an appropriate reason.

Recommended Model: Spiral Model (or Iterative Enhancement / RAD under explicit conditions)

Contextual Assumptions and Engineering Justification: - The problem statement does not provide detailed operational parameters; therefore, the recommended model depends on the project's risk profile, scale, and integration complexity.

- **Primary Recommendation (Spiral Model — Complex / High-Risk Assumption): If the hospital management system is large-scale, unprecedented for the organization, or encompasses mission-critical clinical workflows** (such as real-time patient monitoring, pharmaceutical management, and multi-department database synchronization across multiple locations):

- **The Spiral Model is best suited because it is explicitly risk-driven. Alternatives are evaluated based on risk perceptions in Quadrant 2, and targeted strategies (such as prototyping, simulation, and benchmarking) are deployed in Quadrant 3 to resolve complex technical, operational, and clinical risks before large-scale system construction.**

- **Viable Alternative Recommendation (Iterative Enhancement — Phased Business Delivery Assumption): If the hospital requires routine administrative automation** (such as patient admissions, room allocation, and billing) where requirements are generally understood but resources are phased:

- **The Iterative Enhancement model is appropriate because an administrative core product can be delivered first**, followed by subsequent increments for pharmacy, laboratory, and inventory management based on operational feedback.

- **Viable Alternative Recommendation (RAD Model — Modular / Time-Critical Assumption): If the system can be cleanly partitioned into independent modular subsystems** (e.g., front-desk, pharmacy, billing) to be built by multiple experienced teams using pre-existing components with active staff participation within 60–90 days.

> **ملاحظة الطالب والفخ الامتحاني:** - نص السؤال لم يحدد تفاصيل المستشفى، لذلك التبرير يعتمد على الافتراض الهندسي: - إذا افترضنا نظاماً ضخماً عالي المخاطر ومتعدد الأقسام ← النموذج الحلزوني (Spiral Model) لإدارة المخاطر. - إذا افترضنا أتمتة إدارية روتينية تطلق نواة أساسية ثم تتوسع تدريجياً ← نموذج التحسين التكراري (Iterative Enhancement). - إذا كان النظام قابلاً للتقسيم إلى موديولات مستقلة وفرق متعددة ← نموذج RAD. اذكر الافتراض بوضوح في جواب الامتحان.

Sources: B.B. Agarwal et al. (2010), §2.4, pp. 44–46; §2.6, pp. 47–48; §2.7, pp. 49–50; §2.8, pp. 50–51 (Table 2.2); K.K. Aggarwal & Singh (2007) Slides 37– 40.

Decision Matrix Summary: The Four Selection Pillars

(Cross-Reference from K.K. Aggarwal & Yogesh Singh, Slides 37–40)

When justifying model selection in an examination, always ground the decision in one or more of the four authoritative source matrices:

Characteristics of Requirements:
Understandable and defined early → **Waterfall / RAD**
Changing frequently → **Prototyping / Spiral**
High complexity → **Prototyping / Iterative / Evolutionary / Spiral**

Status of Development Team:
Less experience on similar projects → **Prototyping / Spiral**
New to technology → **Waterfall / Iterative / Evolutionary / Spiral**
Unfamiliar with development tools → **Waterfall / Spiral**

User Involvement:
Active involvement across all phases → **Prototyping / RAD**
Limited user participation available → **Waterfall / Iterative / Evolutionary / Spiral**

Project Type and Associated Risk:
Enhancement of existing system → **Iterative Enhancement / Evolutionary / RAD**
High reliability requirements → **Iterative / Evolutionary / Spiral**
Tight project schedule → **RAD / Iterative / Evolutionary**
Severe resource scarcity → **Prototyping / Spiral**

> **ملاحظة الطالب والفخ الامتحاني:** - إذا طُلب منك تبرير اختيار نموذج في الامتحان، لا تكتفِ بذكر اسم النموذج، بل اذكر المعيار (المتطلبات، المخاطرة، الفريق، مشاركة المستخدم) واستشهد بالصفات المطابقة له.

Sources: K.K. Aggarwal & Singh (2007) Slides 37–40; B.B. Agarwal et al. (2010), §2.8, pp. 50–51.

Retrieval set — Focused Active Recall Drill

1 **Under what explicit condition does the Waterfall model fail for compiler development, and which model replaces it?**

The Waterfall model fails if the programming language grammar, syntax, and type system are experimental or evolving during development. Mid-stream modifications to grammar break downstream parser tables and AST generators across the pipeline, requiring an Evolutionary or Incremental development approach to evaluate syntax iteratively.

Source: K.K. Aggarwal & Singh (2007) Slide 37; B.B. Agarwal et al. (2010), §2.2, p. 38.

2 **Why does the RAD model fail when applied to pioneering or unproven technologies?**

Because RAD relies on accelerated 60–90 day delivery cycles using established high-level visual tools, automated code generators, and well-understood reusable components. Unproven technologies introduce high technical risk, performance uncertainties, and integration hurdles that cannot be accommodated within rigid, compressed timeboxes.

Source: K.K. Aggarwal & Singh (2007) Slide 12, Slide 38; B.B. Agarwal et al. (2010), §2.7, p. 50.